

PrintNet – Assignment #4 (Updated): User Stories & Design Diagrams

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Design D0 — System Context

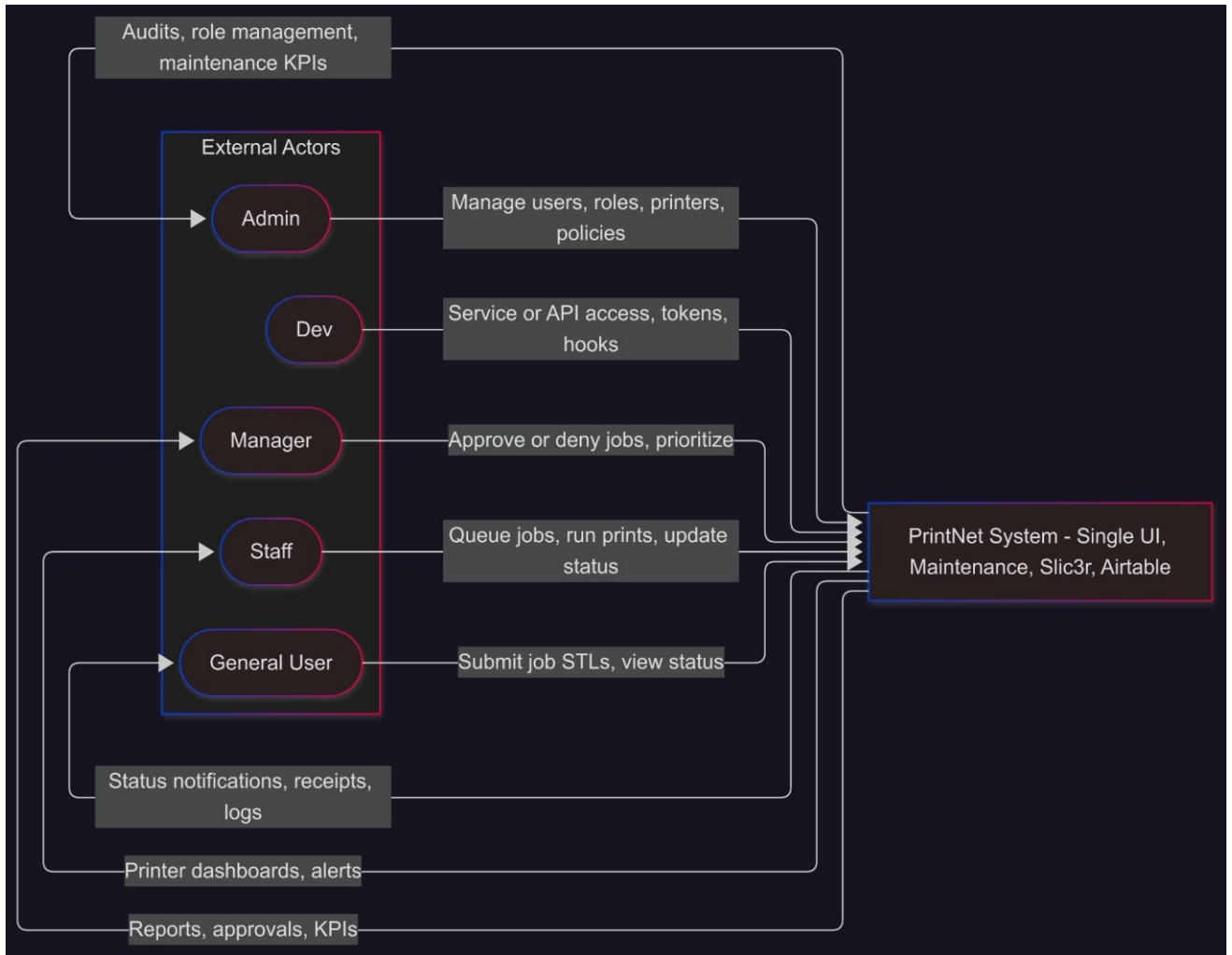


Figure 1. D0 – System Context for PrintNet.

What this shows: who sends inputs to PrintNet (submissions, approvals, admin changes) and what outputs they get back (statuses, dashboards, reports, audits).

Design D1 — Major Subsystems & Data Flow

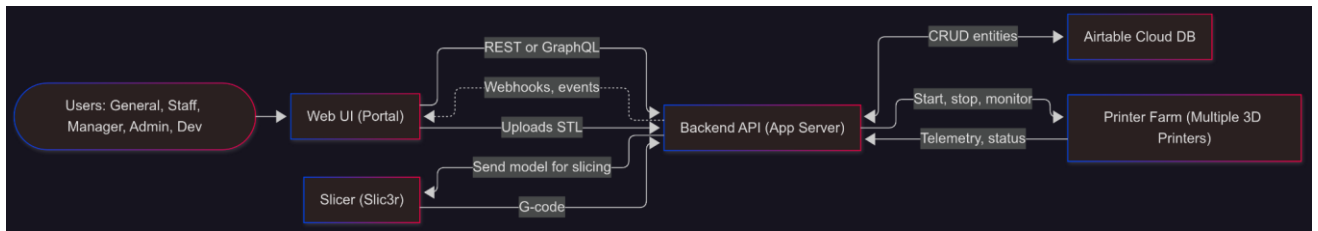


Figure 2. D1 – Major Subsystems and Data Flow.

Inputs/Outputs focus:

- **UI $\rightleftharpoons$ API:** user actions and responses.
- **API $\rightleftharpoons$ DB:** job/user/maintenance records (read/write).
- **API $\rightleftharpoons$ Slicer:** model in $\rightarrow$ G-code out.
- **API $\rightleftharpoons$ Printers:** commands out $\rightarrow$ telemetry/status in.
- **API $\rightarrow$ UI (events):** async notifications.

Design D2 — Detailed Internal Design

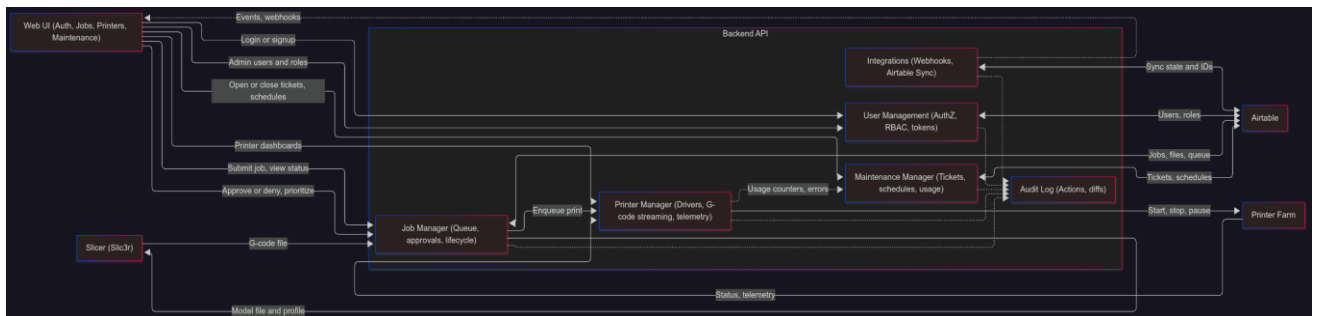


Figure 3. D2 – Detailed Backend and Integrations.

Inputs/Outputs focus:

- **UM** takes auth inputs from UI; outputs role-gated access; persists to DB; audits to AU.
- **JM** takes submissions/approvals; outputs queue state; persists to DB; sends model to **SL** (input) and gets **G-code** (output); enqueues to **PM**.

- **PM** outputs printer commands; inputs printer telemetry; forwards usage/errors to **MM**.
- **MM** inputs maintenance events; outputs tickets/schedule updates to DB.
- **IN** pushes events/webhooks to UI; syncs with Airtable.

ERD — Entities & Relationships

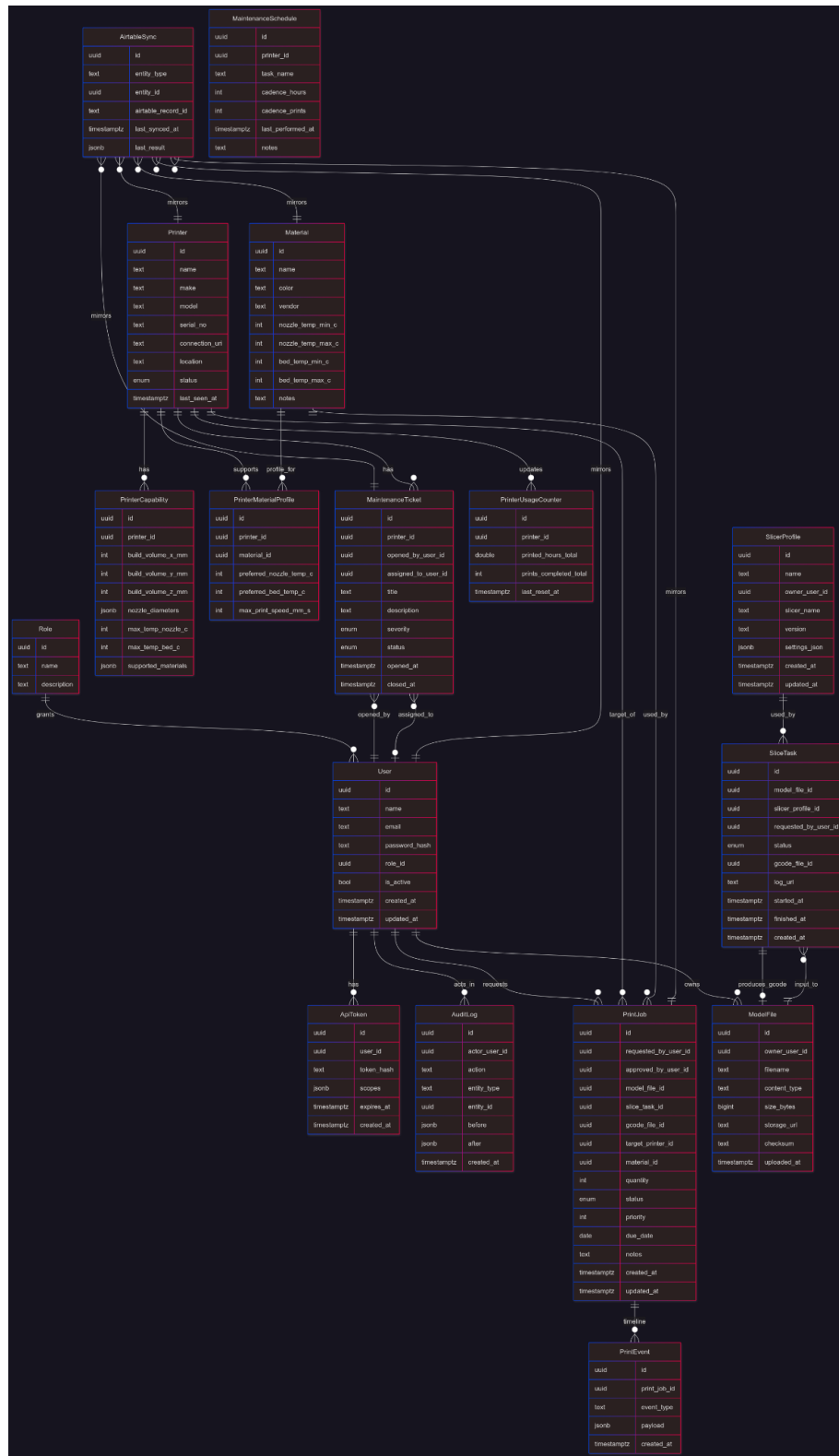


Figure 4. ERD – Data Model for PrintNet.